. **O** curba spline cubica, tensionata este definita pentru o functie f pe intervalul $[1,3]$ prin:

$$S(x) = \begin{cases} S_0(x) = 3(x-1) + 2(x-1)^2 - (x-1)^3 & \text{daca } 1 \le x < 2, \\ S_1(x) = a + b(x-2) + c(x-2)^2 + d(x-2)^3 & \text{daca } 2 \le x < 3. \end{cases}$$

Stiind ca $f'(1) = f'(3)$, determinati a, b, c si d. Estimati $f''(1)$ si $f''(3)$.

Spline – interpolare cu funcții clasa C_1 – f, f''

interpolare — " — C_2 – f (și relații cu f', f'')

- $n+1$: relații inițiale

$S_i(x_i) = f(x_i)$

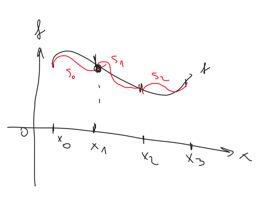
$S_{n-1}(x_n) = f(x_n)$

- $3n-3$ relații de continuitate:

$S_i(x_{i+1}) = S_{i+1}(x_{i+1})$

$S_i'(x_{i+1}) = S_{i+1}'(x_{i+1})$

$S_i''(x_{i+1}) = S_{i+1}''(x_{i+1})$



$4n$ necunoscute $\Leftrightarrow$ $4n$ relații

- spline natural

$S_0''(x_0) = 0$

$S_{n-1}''(x_n) = 0$

- spline tensionat

$S_0'(x_0) = f'(x_0)$

$S_{n-1}'(x_n) = f'(x_n)$

. **O** curba spline cubica, tensionata este definita pentru o functie f pe intervalul $[1,3]$ prin:

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Stiind ca $f'(1) = f'(3)$, determinati a, b, c si d. Estimati $f''(1)$ si $f''(3)$.

- $f'(1) = f'(3) \Rightarrow S_0'(1) = S_1'(3)$

$S_0'(1) = f'(1)$

$S_1'(3) = f'(3)$

$S_0'(x) = 3 + 4(x-1) - 3(x-1)^2$

$S_0''(x) = 4 - 6(x-1)$

$S_1'(x) = b + 2c(x-2) + 3d(x-2)^2$

$S_1''(x) = 2c + 6d(x-2)$

$S_0'(1) = 3$

$S_1'(3) = b + 2c + 3d$ $\Big\} \Rightarrow b + 2c + 3d = 3 \Rightarrow d = \dfrac{3 - 2c - b}{3}$

- 3 relații de continuitate:

$S_0(2) = S_1(2)$ | $S_0(2) = 3 + 2 - 1 = 4$ | $S_1(2) = a \Rightarrow \boxed{a = 4}$

$S_0'(2) = S_1'(2)$ | $S_0'(2) = 3 + 4 - 3 = 4$ | $S_1'(2) = b \Rightarrow \boxed{b = 4}$

$S_0''(2) = S_1''(2)$ | $S_0''(2) = 4 - 6 = -2$ | $S_1''(2) = 2c \Rightarrow \boxed{c = -1}$

$d = \dfrac{3 + 2 - 4}{3} = \boxed{\dfrac{1}{3}}$

$a = 4$
$b = 4$
$c = -1$
$d = 1/3$

. **D**eterminati polinomul de gradul doi care aproximeaza cel mai bine setul de puncte $(-1,-2)$, $(0,1)$ si $(1,2)$. Determinati un astfel de polinom care sa treaca prin origine.

$(-1;-2)$
$(0;1)$ $\Rightarrow f(x)$ exact care să treacă prin cele 3 puncte
$(1;2)$

$P_2(x) = ax^2 + bx + c$

- I. Vandermonde
- I. Lagrange
- I. Neville

$P_2(-1) = a(1)^2 + b\cdot(-1) + c = -2$

$P_2(0) = a\cdot 0^2 + b\cdot 0 + c = 1$

$P_2(1) = a\cdot(1)^2 + b\cdot 1 + c = 2$

$$\Rightarrow \begin{cases} a - b + c = -2 \\ \boxed{c = 1} \\ a + b + c = 2 \end{cases}$$

$$\begin{cases} a - b = -3 \\ a + b = 1 \end{cases}$$

$2a = -2 \Rightarrow \boxed{a = -1}$ $\boxed{b = 2}$

$\boxed{P_2(x) = -x^2 + 2x + 1}$

b) $R_2(x) = ax^2 + bx + c$

$R_2(0) = 0 \Rightarrow c = 0$

$R_2(x) = ax^2 + bx$

$(-1;-2)$
$(0;1)$
$(1;2)$

$\Delta = \sum_{i=1}^{3} (y_i - R_2(x_i))^2$ (M. celor mai mici pătrate)

$\Delta = (-2 - R_2(-1))^2 + (1 - R_2(0))^2 + (2 - R_2(1))^2$

$\Delta = (-2 - a + b)^2 + (1 - 0)^2 + (2 - a - b)^2$

$\Delta = (-a + (b-2))^2 + 1 + (-a - (b-2))^2$

$= a^2 - \cancel{2a(b-2)} + (b-2)^2 + 1 + a^2 + \cancel{2a(b-2)} + (b-2)^2$

$= 2a^2 + 2(b-2)^2 + 1$

$\min \Delta(a,b) = 1$ $\begin{cases} a = 0 \\ b = 2 \end{cases} \Rightarrow \boxed{R_2(x) = 2x}$

3. **F**ormula de integrare Gauss-Lobatto este de forma: $\int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}} \approx \sum_{i=0}^{N} A_i f(x_i)$ si este exacta pentru polinoame de grad $2N-1$. Aceasta utilizeaza ca abscise x_i zerourile polinomului $(1-x^2)T_N'(x)$ ($T_N(x)$ - Cebisev). Dezvoltati o formula de integrare de acest tip cu $N = 3$. Calculati apoi $\int_{-1}^{1} \frac{ln(1+x^2)}{\sqrt{1-x^2}}$ cu formula dedusa.

$N = 3 \Rightarrow A_0, A_1, A_2, A_3$

$\int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}} dx \approx A_0 f(x_0) + A_1 f(x_1) + A_2 f(x_2) + A_3 f(x_3)$

$x_i \Rightarrow (1-x^2) T_3'(x) = 0$

$T_{n+1} - 2xT_n + T_{n-1} = 0$, $T_0 = 1$, $T_1 = x$;

$T_2 = 2xT_1 - T_0 = 2x^2 - 1$

$T_3 = 2xT_2 - T_1 = 2x(2x^2 - 1) - x = 4x^3 - 3x$

$T_3'(x) = 12x^2 - 3$

$(1-x^2) T_3'(x) = 0 \Leftrightarrow (1-x^2)\cdot 3(4x^2 - 1) = 0$

$\Rightarrow x_{1,2} = \pm 1$

$x_{3,4} = \pm \frac{1}{2}$

$\int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}} dx \simeq A_0 \cdot f(-1) + A_1 f\left(-\frac{1}{2}\right) + A_2 f\left(\frac{1}{2}\right) + A_3 f(1)$

polinoame de grad maxim $2N - 1 = 5$ este exactă

$f(x) = P_5(x)$

$\int_{-1}^{1} \frac{P_5(x)}{\sqrt{1-x^2}} dx = A_0 \cdot f(-1) + \dots$

- $f(x) = P_5(x) = 1 \Rightarrow \int_{-1}^{1} \frac{1}{\sqrt{1-x^2}} dx = A_0 \cdot f(-1) + A_1 f\left(-\frac{1}{2}\right) + A_2 f\left(\frac{1}{2}\right) + A_3 \cdot f(1)$

$\arcsin x \Big|_{-1}^{1} = A_0 + A_1 + A_2 + A_3$

$\frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \boxed{\pi = A_0 + A_1 + A_2 + A_3}$

- $P_5(x) = x \Rightarrow -\int_{-1}^{1} \frac{-2x}{2\sqrt{1-x^2}} dx = A_0 \cdot f(-1) + \dots$

$-\sqrt{1-x^2}\Big|_{-1}^{1} = \boxed{A_0 \cdot (-1) + A_1 \cdot \left(-\frac{1}{2}\right) + A_2 \cdot \left(\frac{1}{2}\right) + A_3 = 0}$

$-0 - (-0) =$

$\left((1-x^2)^{\frac{1}{2}}\right)' = \frac{1}{2} \cdot \frac{-2x}{\sqrt{1-x^2}}$

- $P_5(x) = x^2 \Rightarrow \int_{-1}^{1} \frac{x^2}{\sqrt{1-x^2}} dx = A_0 \cdot f(-1) + A_1 \cdot f\left(-\frac{1}{2}\right) + A_2 \cdot f\left(\frac{1}{2}\right) + A_3 \cdot f(1)$

$i = \int_{-1}^{1} \frac{x^2}{\sqrt{1-x^2}} dx = \int_{-1}^{1} \frac{\overset{f \cdot g'}{x \cdot x}}{\sqrt{1-x^2}} dx = \overset{f \cdot g}{-\sqrt{1-x^2} \cdot x}\Big|_{-1}^{1} - \int_{-1}^{1} \overset{f' \cdot g}{-\sqrt{1-x^2}} dx$

$f = x$ $\quad g = -\sqrt{1-x^2}$

$= 0 + \int_{-1}^{1} \frac{1 - x^2}{\sqrt{1-x^2}} dx$

$= \int_{-1}^{1} \frac{1}{\sqrt{1-x^2}} dx - i$

$2i = \int_{-1}^{1} \frac{1}{\sqrt{1-x^2}} dx = \pi \Rightarrow i = \frac{\pi}{2}$

$\frac{\pi}{2} = A_0 \cdot (-1)^2 + A_1 \cdot \left(-\frac{1}{2}\right)^2 + A_2 \cdot \left(\frac{1}{2}\right)^2 + A_3 \cdot 1^2$

$\boxed{\frac{\pi}{2} = A_0 + \frac{1}{4} A_1 + \frac{1}{4} A_2 + A_3}$ (3)

- $P_5(x) = x^3 \Rightarrow \int_{-1}^{1} \frac{x^3}{\sqrt{1-x^2}} dx = A_0 \cdot f(-1) + A_1 \cdot f\left(-\frac{1}{2}\right) + \dots$

$\int_{-1}^{1} \frac{x^3}{\sqrt{1-x^2}} = 0 = A_0 \cdot (-1)^3 + A_1 \left(-\frac{1}{2}\right)^3 + A_2 \cdot \left(\frac{1}{2}\right)^3 + A_3 \cdot 1^3$

$$\begin{cases} A_0 + A_1 + A_2 + A_3 = \pi \\ -A_0 - \frac{1}{2}A_1 + \frac{1}{2}A_2 + A_3 = 0 & (2) \\ A_0 + \frac{1}{4}A_1 + \frac{1}{4}A_2 + A_3 = \frac{\pi}{2} \\ -A_0 - \frac{1}{8}A_1 + \frac{1}{8}A_2 + A_3 = 0 & (4) \end{cases}$$

$\xrightarrow{(4)+(4)} \boxed{\begin{matrix} A_0 = A_3 \\ A_1 = A_2 \end{matrix}}$

$\Rightarrow \begin{cases} 2A_0 + 2A_1 = \pi \\ 2A_0 + \frac{1}{2}A_1 = \frac{\pi}{2} \end{cases} \Rightarrow \begin{cases} A_0 + A_1 = \frac{\pi}{2} \\ A_0 + \frac{1}{4}A_1 = \frac{\pi}{4} \end{cases}$

$\frac{3}{4}A_1 = \frac{\pi}{4}$

$\Rightarrow \boxed{A_1 = \frac{\pi}{3};\ A_0 = \frac{\pi}{6}}$

$\boxed{\int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}} dx \simeq \frac{\pi}{6} \cdot f(-1) + \frac{\pi}{3} f\left(-\frac{1}{2}\right) + \frac{\pi}{3} f\left(\frac{1}{2}\right) + \frac{\pi}{6} f(1)}$

b) $\int_{-1}^{1} \frac{\ln(1+x^2)}{\sqrt{1-x^2}} dx \simeq \frac{\pi}{6} \cdot \ln(1+1) + \frac{\pi}{3} \cdot \ln(1 + 1/4) + \frac{\pi}{3} \ln(1 + 1/4) + \frac{\pi}{6} \ln(1+1)$

$= \boxed{\frac{\pi}{3} \ln 2 + \frac{2\pi}{3} \ln \frac{5}{4}}$

4. **C**alculati coeficientii metodelor de tip Adams: (M) Milne: $y_{k+1} = y_{k-3} + h \sum_{i=0}^{2} m_i f_{k-i}$ si (S) Simpson: $y_{k+1} = y_{k-1} + h \sum_{i=-1}^{1} s_i f_{k-i}$. Stabiliti gradul de valabilitate al fiecareia. Definiti cu cele doua formule o metoda Predictor-Corector.

Metode implicite: – Adams – Moulton:
$\downarrow$
corectorul $\quad y_{k+1} = y_k + h \sum_{j=0}^{2} d_j \nabla^j f_{k+1}$

Metode explicite – Adams – Bashforth:
$\downarrow$
predictorul

Utilizând în metoda implicită aproximația făcută la metoda explicită se obține met. predictor corector

Milne. Se consideră setul de puncte

$y_{k+1} = y_{k-3} + h(m_0 f_k + m_1 f_{k-1} + m_2 f_{k-2})$

$h = 1$
$x_{k-3} = 0$
$x_{k-2} = 1$
$x_{k-1} = 2$
$x_k = 3$
$x_{k+1} = 4$

$f_k = y'(x_k)$

$y = 1 \Rightarrow f = y' = 0 \Rightarrow 1 = 1$

$y = x \Rightarrow f = y' = 1 \Rightarrow x_{k+1} = x_{k-3} + (m_0 \cdot 1 + m_1 \cdot 1 + m_2 \cdot 1)$

$\boxed{4 = 0 + m_0 + m_1 + m_2}$

$y = x^2 \Rightarrow f = y' = 2x \Rightarrow x_{k+1}^2 = x_{k-3}^2 + 2(m_0 \cdot x_k + m_1 \cdot x_{k-1} + m_2 \cdot x_{k-2})$

$\boxed{16 = 0 + 2(3m_0 + 2m_1 + m_2)}$

$\boxed{y = x^3} \Rightarrow f = y' = 3x^2 \Rightarrow x_{k+1}^3 = x_{k-3}^3 + 3(m_0 \cdot x_k^2 + m_1 \cdot x_{k-1}^2 + m_2 \cdot x_{k-2}^2)$

$64 = 0 + 3 \cdot (9m_0 + 4m_1 + m_2)$

$$\begin{cases} 9m_0 + 4m_1 + m_2 = 64/3 \\ 3m_0 + 2m_1 + m_2 = 8 \\ m_0 + m_1 + m_2 = 4 \mid \cdot 2 \end{cases}$$

$2m_0 + 2m_1 + 2m_2 = 8$

$m_0 - m_2 = 0 \Rightarrow m_0 = m_2$

$\Rightarrow \begin{cases} 10m_0 + 4m_1 = 64/3 \\ 4m_0 + 2m_1 = 8 \mid \cdot 2 \end{cases}$

$2m_0 = 64/3 - 16 = \frac{64 - 48}{3} = \frac{16}{3}$

$\boxed{\begin{matrix} m_0 = m_2 = \frac{8}{3} \\ m_1 = 4 - 2 \cdot \frac{8}{3} = -\frac{4}{3} \end{matrix}}$

- Grad valabilitate $= 3$ = gradul maxim al polinomului care atunci când este înlocuit în locul lui y, formula este exactă

b) Formula explicită furnizează predicţia:

$$y_{k+1}^{(p)} = y_{k-1} + \frac{h}{3}(7f_k - 2f_{k-1} + f_{k-2})$$

iar formula implicită corectează această valoare, astfel:

$$y_{k+1}^{(c)} = y_{k-1} + \frac{h}{3}[f(x_{k+1}, y_{k+1}^{(p)}) + 4f_k + f_{k-1}]$$

M: $y_{k+1}^{(p)} = y_{k-3} + h(m_0 f_k + m_1 f_{k-1} + m_2 f_{k-2})$

S: $y_{k+1}^{(c)} = y_{k-1} + h(S_{-1} f_{k+1} + S_0 f_k + S_1 f_{k-1})$

$y_{k+1}^{(p)} = y_{k-3} + h(m_0 f_k + m_1 f_{k-1} + m_2 f_{k-2})$

$y_{k+1}^{(c)} = y_{k-1} + h(S_{-1} \cdot f(x_{k+1}, y_{k+1}^{(p)}) + S_0 f_k + S_1 f_{k-1})$

. **S**crieti o functie MATLAB care implementeaza o metoda RK2 (cu $\alpha = 1$) pentru rezolvarea ecuatiei $y'' = k^2 y$, cu $y(0) = \frac{1}{k}$ si $y'(0) = -1$, pe intervalul $[0,a]$, $a > 0$.

① Pentru $y' = f(t,y)$ $\quad t_0 = 0$

$y' = ky$ $\quad y(0) = 0$

$t_{i0} = t_0 + i \cdot h$

RK2: $\begin{cases} y_{i0} = y_i \\ y_{i1} = y_i + h \cdot \alpha \cdot f(t_{i0}, y_{i0}) \\ y_{i+1} = y_i + h\left(1 - \frac{1}{2\alpha}\right) \cdot f(t_{i0}, y_{i0}) + h \cdot \frac{1}{2\alpha} \cdot f(t_{i0}, y_{i1}) \end{cases}$

metoda Euler–Cauchy ($\alpha = 1$) – RK2

$y_{i+1} = y_i + \frac{h}{2}\left(f(t_{i0}, y_{i0}) + f(t_{i0}, y_{i1})\right)$

Pentru ec. dif ordin 2:

$g = y'' = ay' + by \rightarrow \begin{cases} y_{i+1} \\ y'_{i+1} \end{cases}$ $\quad h = y'' - ay' = by$

$$\begin{cases} y'_{i+1} = y'_i + \frac{h}{2}\left(g(t_{i0}, y_{i0}, y'_{i0}) + g(t_{i0}, y_{i1}, y'_{i1})\right) \\ y_{i+1} = y_i + \frac{h}{2}\left(y'_{i0} + y'_{i1}\right) \end{cases}$$

$$\begin{cases} y_{i0} = y_i \\ y'_{i0} = y'_i \\ y_{i1} = y_i + h \cdot \left(\frac{y_{i0} + y'_{i1}}{2}\right) \\ y'_{i1} = y'_i + h \cdot g(t_{i0}, y_{i0}, y'_{i0}) \end{cases}$$ (RK – Nyström)

```
function [y,dy] = RK(g,t0,y0,dy0, h ,n)
  y=zeros(n+1,1);
  dy=zeros(n+1,1);

  y(1)= y0;
  dy(1)=dy0;
  for  i = 1:n
     tio= to+ dt*(i-1)
     yio= y(i)
     dyio =dy(i)
```

$t_0 = 0$

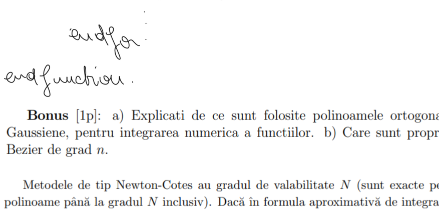
$y_0 = y(t_0) = 1/k$

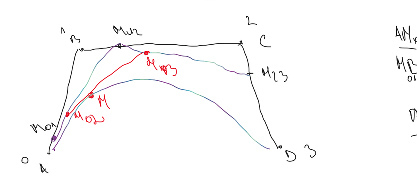
$dy_0 = y'_0 = y'(t_0) = -1$

y – $n+1$ puncte

dy – derivatele

[illegible]



[illegible]